

Design, development, and analysis of validity for a computer-based cognitive rehabilitation (CBCR) program for children with specific learning disorder (SLD) – Dyslexia: A design-based approach

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ABSTRACT

Introduction: Dyslexia is a prevalent learning disorder that impairs children's reading abilities and language processing. Cognitive deficits, particularly in language processing and phonological awareness, pose significant challenges, making targeted interventions essential for improving reading skills and academic outcomes.

Objective: This study aimed to develop a computer-based cognitive rehabilitation program specifically tailored for children with dyslexia, designed to enhance cognitive and academic skills through targeted exercises and adaptive training.

Methods: A developmental approach was used to construct the program, incorporating insights from prior research and the needs of dyslexic children. The program included practical game tasks addressing key cognitive deficits. Validity was assessed through semi-structured interviews with professionals, using both qualitative and quantitative feedback. K-means clustering was applied to categorize tasks based on cognitive functions such as executive functions, memory, and attention, aligning tasks with specific cognitive goals.

Results: Expert feedback led to refinements, achieving a content validity coefficient of 0.74, with indices for simplicity, clarity, and relevance at 0.93, 0.97, and 0.94, respectively.

Conclusion: The program effectively addresses cognitive deficits in dyslexia through adaptive training and clustered tasks. While expert evaluations are promising, further research is needed to validate its effectiveness in real-life settings and for diverse populations.

1. Introduction

Specific learning disorder (SLD)¹ is one of the most common neurodevelopmental disorders [1]. It is a disorder in which educational skills are significantly weaker than is expected for each age group and it is not caused by a lower Intelligence Quotient (IQ) [2]. Although it can be detected in young students, in some instances, it may not be identified until a later stage in life, impacting daily activities and academic and professional achievement [3]. The school-aged children diagnosed with this disorder are estimated to be between 5 and 15 % of the total population [4]. A persistent impairment in at least one of the three main areas of learning viz., reading difficulties (dyslexia), written expression (dysgraphia), and mathematics calculation (dyscalculia) are specific

types of problems of individuals with SLD [5].

The key weak cognitive functions in the context of SLD include executive functioning, coordination, and attention [6], and memory, especially working memory [7,8]. Moreover, deficits in executive functioning include trouble with organizing, planning, remembering specifics, forming strategies, and effectively allocating time and space [9,10]. The inability to access information processing due to weak cognitive functions can lead to frustration, low self-esteem, and social withdrawal [11]. Additionally, children's deficiencies in visual abilities [12] and auditory skills [13] may make SLD more serious, especially when it comes to visual attention impairments [14] and perceptual problems [15].

Approximately 80 % of all learning disorders are dyslexic, making it

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¹ Specific learning disorder (SLD).

the most common and well-defined type of learning disability [16]. In particular, dyslexia is characterized by problems in learning and processing language, which can lead to difficulties with writing, spelling, and reading [17]. For instance, people suffering from dyslexia, frequently have trouble connecting sounds on a page to letters [18]. Understanding how these weaknesses affect people with dyslexic disorders can help researchers plan for more effective training and rehabilitation programs [19]. Under these conditions, support services are essential for learners to adapt to the severity of the problem and maximize their learning potential.

In the absence of additional support, individuals with dyslexic disorders commonly exhibit poor academic performance [20] which is primarily due to challenges in organizing data [21]. Given these considerations, researchers have the opportunity to develop more tailored training programs by exploring how these areas of weakness impact individuals with dyslexic disorders. Such attempts can facilitate more effective rehabilitation instruction and support [19]. In recent studies, the phrase “cognitive functions,” notably “executive functions,” has been used to refer to a wide variety of cognitive processes including inhibitory control, planning, working memory, and cognitive flexibility [22]. Indeed, enhancing cognitive functions enables dyslexic children to address their cognitive deficits [23], improve psychological well-being [24], and mitigate overall educational problems [25].

Certainly, individuals with learning disorders can harness the inherent potential of these cognitive processes to organize activities, accomplish tasks, and attain their goals [26]. Peters et al. [27] revealed that visual perceptual training improves reading skills in dyslexic individuals, resulting in improved fluency, comprehension, and accuracy. Similarly, research by Caldani et al. [4] found that short visual attentional training enhances dyslexic children’s reading speed by reducing the duration of fixations. Therefore, cognitive interventions and treatments have emerged and are known as popular methods for enhancing cognitive abilities [28]. Advances in neuroscience have also demonstrated the brain’s unique characteristic of “neuroplasticity” or “neural plasticity” [29], which can enhance the brain’s cognitive processes at any age, leading to improving cognitive functioning [30].

Cognitive rehabilitation is a technique for restoring or improving impaired cognitive abilities through training and the provision of intentional stimuli regarding improving cognitive functions for enhancing an individual’s capability to perform tasks and activities [31]. This process consists of a range of evidence-based interventions that are used to help persons with brain damage or other forms of cognitive impairment recover or compensate for cognitive deficiencies [32]. The Commission on Accreditation of Rehabilitation Facilities (CARF) defines cognitive rehabilitation as “performance-oriented therapy activities aimed at reinforcing prior behavioral patterns while establishing new ones.” It involves compensating for the impaired functioning of the nervous system by engaging in activities or providing cognitive mechanisms [33]. Taking this strategy, the therapist evaluates the information gathered from the session’s evaluations and creates tasks to strengthen and reconstruct the brain’s cognitive functions [34]. Research revealed that even in healthy people, an increase in cortical thickness has been documented with participating in cognitive training programs [35].

In general, both traditional and computer-based cognitive rehabilitation interventions have the potential to enhance cognitive capacity in individuals with and without cognitive issues [36]. However, computer-based cognitive rehabilitation is now an emerging approach for many intervention programs [32], which plays a complementary role to the traditional rehabilitation method. In the area of education, computer-based cognitive rehabilitation offers numerous benefits, particularly for students facing dyslexic problems [37]. According to certain studies, computer-based cognitive rehabilitation games even have exhibited greater effectiveness in enhancing performance compared to conventional rehabilitation approaches [38,39]. The features of a computer-based intervention program encompass attributes such as sound, color,

movement, and instant feedback. In addition, they can contain more adaptability, flexibility, and personalization tailored to the characteristics of the participants which potentially can speed up the recovery of the respondents by providing a range of services [40]. Furthermore, incorporating pseudo-game elements into these programs enhances training perseverance and motivation [41], resulting in the development of a positive mindset among these students [42].

2. Related research / background

2.1. Key cognitive function impairments in dyslexic children

The development of cognition in children is a crucial aspect of their overall growth and learning, and encompasses the acquisition, organization, and utilization of mental information, with associated skills such as language, visual construction, perception, memory, attention, and executive processes [43–45]. Cognitive interventions can be considered as a necessary tool to enhance the cognitive weaknesses of dyslexic children, therefore for developing any intervention program first it needs cognitive processes recognition. Traditionally, this disorder is recognized based on criteria such as phonological awareness, processing, and reading fluency, but recently it has been proven that it affects cognitive aspects beyond language-specific domains [46]. Recent research suggests that individuals with dyslexia often exhibit deficits in universal information processing systems, affecting attention, visual working memory, naming, visual perception, and speed of processing, as well as executive functions like inhibition, shifting, and planning [6,47,48]. These cognitive impairments are strongly associated with key reading skills such as phonological decoding, orthographic processes, and fluency measures [49].

As previously stated, one of the fundamental deficiencies of dyslexic children is visual processing and scanning weaknesses that can be effective in proper reading and have a fundamental function in decoding [50]. Indeed, phonological decoding requires the visual-spatial segmentation of a string of letters into grapheme, and also a visual search activity in the early stages of learning [51]. Readers should first pay attention to single letters in sequence to recognize the proper word. According to Stevens et al. [52], reading requires visuospatial attention, especially in the early phases of learning. Letters must be carefully selected through quick visual attention orienting at the start of reading acquisition. As a result, visuospatial attention and phonological skills are essential for correct reading [53]. According to Stefanac et al. [54], visual attention is even more significant than visual working memory capacity, indicating that a deficit in visual attention is a well-documented factor that contributes to dyslexia [55]. The National Center for Learning Disabilities breaks up this concept into the following sections: visual discrimination, visual figure-ground discrimination, visual sequencing, visual-motor processing, visual memory, visual closure, and spatial relationships [56].

In addition, supervisory attention is one of the most significant cognitive abilities that facilitate language learning [57]. It includes focused attention (maintaining focus over time and inhibiting what is irrelevant) as well as divided and switching attention (flexibly switching focus from one relevant category to another). This attention for inhibition underlies these two kinds of attention during rapid automatic naming of alternating letters and numerals, and self-monitoring (repetitions during word-finding for spellings based on initial letter). When it comes to attention deficits in dyslexic children, divided attention and focused attention are more noticeable. The flexibility of attention is required for shifting focus from word to word when reading, as well as the ability to pay attention to multiple things at once (divided attention), as reading in dyslexia disorder, is not natural and must be intentionally managed [58]. As a result of focused attention, active visual spatial attention, and other cognitive skills like planning are skills that help children enhance their reading skills, both in speed and accuracy, by allowing them to access a more automatic style of reading. Both kinds of

the mentioned attention play a crucial role in acquiring reading skills in the end, but in this way, their contribution may be changed over time to achieve this goal. For instance, selective visuospatial attention is no longer involved in the decoding process in the third class, and focused attention appears to be more involved in rapidity [59,60].

Moreover, working memory is the most essential cognitive component for proficient reading after phonological processing and during reading. It is noticeable that all parts of working memory, for instance, phonological, visual-spatial, and executive are important in the process of cognition development [61]. The processing dimension of working memory is named executive working memory, also known as a central executive function which is a higher-level component, it is in charge of these storage components and it includes critical activities like inhibition, switching, and updating [62]. Working memory problems, in particular, are currently considered one of the key decaying features of dyslexia [63]. Provazza et al. [64] investigated visual working memory in adult English dyslexics and found that two out of three tasks related to visual working memory, namely reading decoding and fluency, were impaired in the dyslexic group. These impairments were associated with deficits in phonological and visual-spatial short-term memory. However, they observed that reading comprehension was linked to executive working memory [65].

Additional important cognitive components within the executive function of children with dyslexia include response time in coordination, visual scanning in perception, planning and processing speed in reasoning, and naming in language. Response time is associated with coordination, while reasoning represents a high-level component of general intelligence. Assessments of reasoning and oral language proficiency can provide a more accurate indication of academic potential by removing tests involving lower-level, less complex abilities, which are frequently the source of a reading impairment [66]. Dyslexic students' reading and spelling problems are usually caused by cognitive and language deficits. Phonological awareness, orthographic awareness, memory, rapid automatic naming, and processing speed are some of the factors that may play a role in dyslexia problems [67].

Difficulties with processing speed are linked to problems in reading accuracy and reading rate. Processing speed includes the rate of perception (input), output (such as vocal or motor responses), or the speed at which perceptual, cognitive, and output processes are integrated [67]. The ability to perform cognitive activities automatically and fluently is measured by processing speed [68]. Students with dyslexia usually have processing speed impairments in both language and nonlinguistic activities [69]. Reading accuracy, reading rate, and reading comprehension problems can all be enhanced by a rapid automatized naming (RAN) problem [70], which is the ability to quickly

retrieve the names of familiar things or symbols [71]. School psychologists can use RAN defects to distinguish students with dyslexia from pupils with other types of learning difficulties [72]. These cognitive impairments for people with dyslexic disorders typically can be shown in a hierarchical structure, shown in Fig. 2, with a detailed description of each component.

2.2. The effects of computer-based cognitive rehabilitation approaches on cognitive functions

Several research studies have indicated that computer-based cognitive rehabilitation methods yield diverse and significant impacts on cognitive abilities. These interventions, often entirely digital, utilize automated customization programs to efficiently modify the complexity, duration, and pace of stimulus in a presentation to suit individual needs [73,74]. This adaptability guarantees that interventions are precisely tailored to users' unique cognitive profiles, thereby optimizing their effectiveness and relevance. Furthermore, research has demonstrated notable improvements in response to inhibition and executive functions following computer-based cognitive training [75]. These enhancements are essential for tasks requiring self-regulation and decision-making, contributing to overall better cognitive function and adaptability. Additionally, increases in reasoning, logic, and cognitive flexibility have been reported, alongside improvements in auditory and visual processing [76]. These findings emphasize the broad impact of computer-based interventions on cognitive processing and problem-solving abilities.

Some studies have also indicated enhancements in information processing speed as a crucial factor for task completion [77]. Moreover, reductions in cognitive avoidance and cognitive failure have been recorded, suggesting advancements in cognitive resilience and adaptability [78]. Notable improvements have also been observed in memory function, including working memory, short-term memory, and long-term memory [79]. These enhancements are fundamental for learning, retention, and cognitive performance across various tasks and domains. To be more precise, computer-based cognitive training enhances attentional abilities and span [75,80] resulting in the facilitation of task engagement, learning, and information processing as well as contributing to overall cognitive efficiency and performance.

Technological advancements present opportunities to support children with SLDs, strengthening cognitive skills essential for reading ability [81]. Moreover, technology-based interventions enhance academic performance, improving reading and math skills [82]. Improvements have also been noted in general intelligence, spelling, arithmetic and reading comprehension, and sign language [76]. These improvements in academic skills highlight the broader impact of computer-

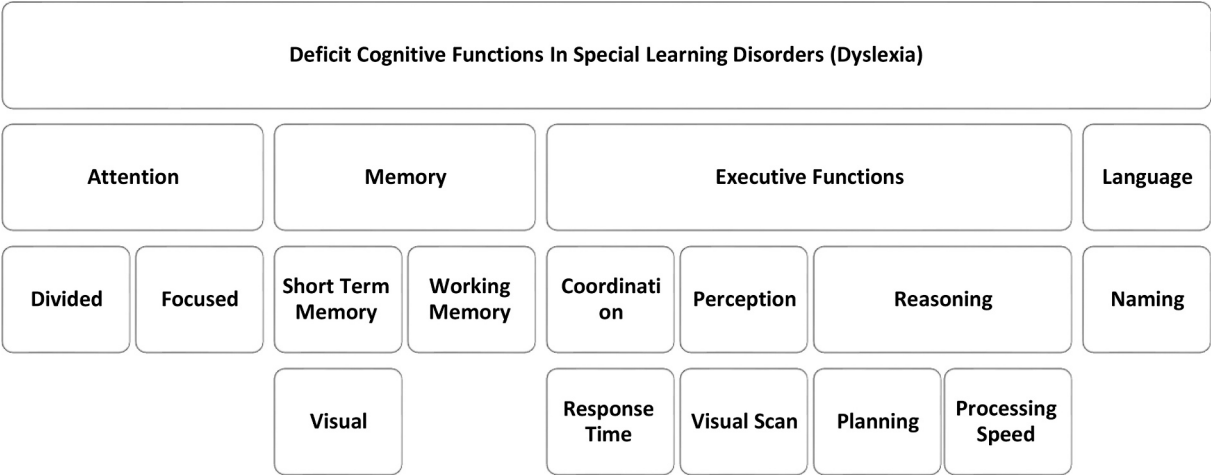


Fig. 1. Key research stages in designing and developing a cognitive game-based rehabilitation program.

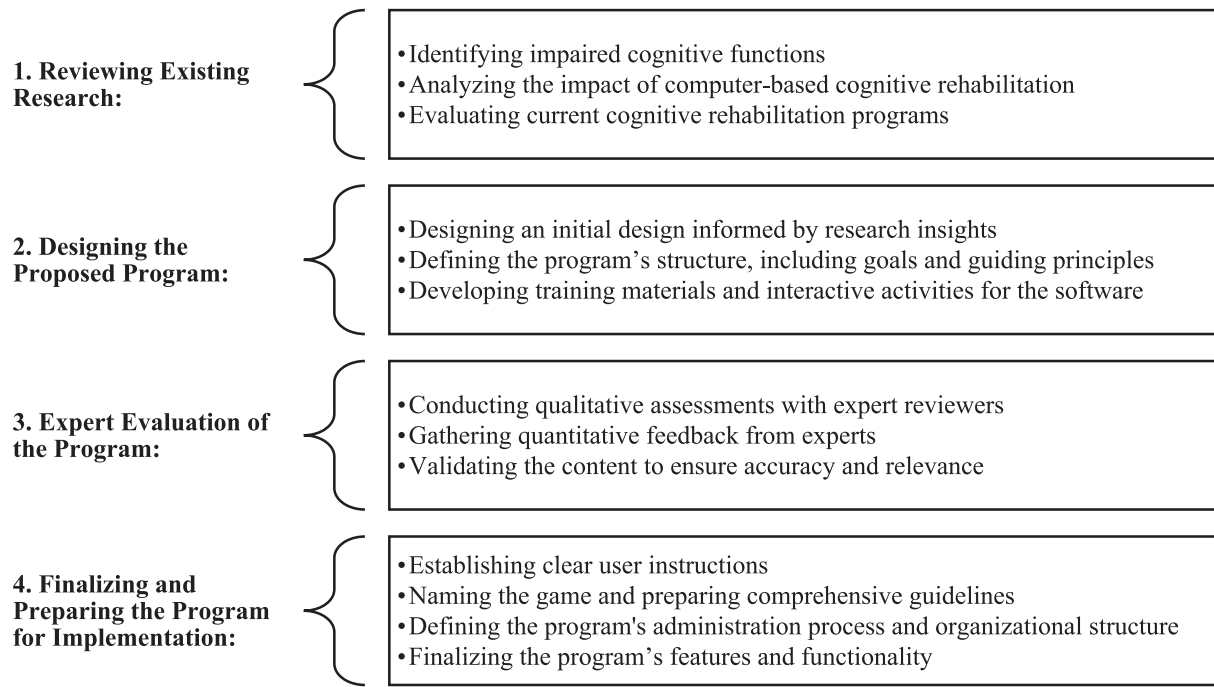


Fig. 2. Deficit cognitive functions in special learning disorders (dyslexia).

based interventions on cognitive development and educational outcomes. Innovative web-based programs offer promising solutions for dyslexia screening, addressing issues of cost and accessibility [83].

Computer-based therapies have also been found to be effective in improving self-control and motivation [40]. Enhanced self-regulation and motivation are vital for consistent involvement in cognitive tasks and learning, fostering enduring cognitive progress and development, as indicated by numerous studies [75]. Based on these findings, the integration of computer-based cognitive rehabilitation techniques exhibits the potential to enhance cognitive functions, improve academic performance, and foster overall cognitive growth. These results underline the capability of computer-based interventions to impact cognitive functions across different domains and environments. To develop effective computer-based rehabilitation programs, careful consideration is crucial to match the program to the individual needs and levels of disability, ensuring effective utilization [84].

As computer-aided cognitive rehabilitation has engaging multimedia features, they are particularly beneficial for children with limited attention spans [85], for instance for their letter recognition skills [86]. Research revealed that educational linguistic games software are better than regular methods at making children understand words and sounds enhancing their both semantic and phonological awareness [87]. Based on estimations, it is indicated that approximately 20 % of primary school students in Iran are affected by learning disabilities [88], underscoring the significance of this matter and the necessity for interventions aimed at treatment and prevention for this group of children.

This study was conducted in response to this important issue, seeking to identify viable solutions. aimed to develop a computer-based cognitive training program tailored for dyslexic children. It considered the prerequisites of the design process, as well as the structural and implementation aspects of such training. By addressing these aspects, the study developed a framework to assist in constructing a cognitive rehabilitation program aimed at enhancing the cognitive skills of dyslexic children. This framework could serve as a valuable resource for researchers and clinical practitioners seeking to develop instructional procedures that enhance cognitive functions as learning facilitators. From an academic viewpoint, it's important to emphasize the novelty of the research and illustrate its relationship to past and potential future

studies.

A gap in existing academic studies initiated this research, as many researchers have looked at the effectiveness of cognitive training programs and interventions and have detailed the contents of these programs, but a few numbers of them have looked at how these programs are designed and developed. The present study aimed to address this gap by examining various steps involved in creating a cognitive computer-based intervention program. First, we attempt to identify key stages of cognitive program development, recognizing their importance in making effective interventions. Second, we try to outline the specific features that need to be considered at each stage, including how to deal with any challenges that might come up. Finally, it is important to know about the specification of one program and how it can be customized to the target group's needs, regarding graphic design, concepts or tasks difficulty, and levels.

We also aim to contribute to this objective and hope our findings will aid future research. Additionally, we focus on the need to have clear procedures and structures for making cognitive interventional programs. These guidelines, if available would be helpful for researchers, giving them a roadmap for designing and using cognitive programs. By advocating for the adoption of these guidelines, we hope to improve the quality of research in this field and make sure that the clients and stakeholders can collaborate effectively and utilize it. Overall, this study was carried out to design an effective and user-friendly computer-based cognitive rehabilitation program tailored for children with dyslexia, addressing specific research objectives. These included identifying essential features necessary for developing an efficient rehabilitation program, examining the effects of computer-based cognitive rehabilitation, investigating recent interventions in Farsi, and delineating the process for creating and validating an innovative computer-based cognitive intervention to empower dyslexic children.

3. Methodology

This study used a descriptive approach to design and evaluate a reliable computer-based cognitive rehabilitation program for children experiencing challenges with dyslexia-related learning difficulties. Previous research has provided specific suggestions for creating and

adjusting cognitive rehabilitation programs. This study aimed to develop a new program by mimicking established methods and following similar steps [89–91]. The research followed Stephan and Stephan [92] theory when developing an evidence-based training and rehabilitation program. The initial stage involves choosing the program's target audience, followed by defining its goals. Then, the theories linked to the target group's cognitive rehabilitation and specific psychological and communicative processes should be explored. Identifying relevant methodologies and training procedures in line with these processes is the subsequent phase of the study. In the final phase, the effectiveness of this program, along with the techniques yielding the best results, must be evaluated. Fig. 1 illustrates the key stages of developing this cognitive rehabilitation program, detailed as follows:

3.1. Research background in psychological and educational fields

The initial step in developing a cognitive rehabilitation program involved a comprehensive review of the literature in the field. During this phase, particular attention was given to studies concerning individuals with learning disabilities, specifically dyslexia, and their challenges. This stage aimed to accurately identify the cognitive skill gaps of these individuals and explore effective strategies to cope with these deficiencies. Consequently, the intervention's foundation was grounded in a thorough examination of prior research on learning disabilities, particularly dyslexia, and cognitive rehabilitation initiatives. This part of the research was conducted to accomplish the following objectives, each of which will be extensively discussed in the findings section:

- A) Recognizing the most important affected domains of cognitive functions in dyslexic children.
- B) Investigating the effects of existing computer-based cognitive rehabilitation methods on cognitive functions.
- C) Developing a computer-based cognitive rehabilitation program for learning disabilities and identifying its characteristics.

3.2. Program development

Using the information and insights gathered in the first phase, which is described in the Background section, this step entailed creating an initial outline for a cognitive rehabilitation program. Subsequently, essential stages and sessions of the rehabilitation program were developed, with each targeting distinct cognitive functions. Drawing from previous research findings and the researcher's recommendations, several program objectives were formulated. In other words, the initial identification of deficiencies across different cognitive functions in these individuals led to the selection of program objectives aimed at enhancing various cognitive skills. Overall, a range of practical game tasks was designed based on prior similar research in cognitive rehabilitation and they are developed to rehabilitate the most impactful cognitive functions observed in children with learning disabilities which was explored in the previous phase. Afterward, the task durations and the program's length and frequency of intervention were established to prepare it as a package and finally, the considerations related to its implementation were decided.

3.3. Evaluation of the program by qualified experts

This step of the research aimed to analyze and assess the validity of the program's tasks using expert judgment regarding the program's goals and components. The goal was to identify and make any essential changes to this cognitive rehabilitation program making it more consistent with the special needs of dyslexic children and their cognitive deficiencies. To validate and improve this rehabilitation program, a criterion-based purposive sampling technique [93] was utilized to select a statistical sample of specialists possessing knowledge and proficiency

in Dyslexia. The specialists needed to have familiarity with the process of guideline development outlined in the APA's evaluation standards before beginning their assessment [94], as it wasn't a good idea to only look at the guidelines without considering how they were made. So, forming a balanced group of experts to give advice could help them thoroughly check important research, treatment standards, costs, benefits, and any conflicts of interest in the study [95].

Accordingly, 16 experienced experts participated as sample participants, guided by predetermined guidelines for the program assessment. The experts involved in the study had varying academic qualifications and expertise in fields such as psychology, special education, computer software engineering, computer-human interaction (HCI), and psychometry. Additionally, they possessed knowledge and experience in clinical rehabilitation and developing computer-based programs for children. Around 81.25 % held doctoral degrees and worked as faculty members at famous universities in Iran, particularly the University of Tehran. The remaining 18.75 % held master's degrees in the fields above and had valuable experience in cognitive rehabilitation for children. Their qualifications are detailed in Table 1.

The initial program iteration comprised 24 games and the assessment questionnaire comprising 18 items for each game was devised to evaluate exercises and rehabilitation sessions against parameters such as the conceptual and theoretical underpinnings of cognitive rehabilitation, as well as the suitability of tasks and activities about the session objectives. It established standards for assessing the alignment of session objectives and content with each task goal in the program. A mixed evaluation approach was employed to enhance the program's utilization of appropriate terminology, placement of features, and establishment of suitable scoring criteria. These aspects were assessed by the experts using two known methods viz., content validity ratio (CVR) and content validity index (CVI) to objectively evaluate content validity [96]. The content validity index was used to evaluate the program's validity, to evaluate the content validity of the program, a combination of qualitative and quantitative approaches was employed to ascertain whether the existing tool adequately encompasses all significant and primary aspects of the concept. Finally, experts were requested to provide feedback about which items should be revised or changed.

3.3.1. Content validity ratio (CVR)

To assess this index, experts were tasked with assigning scores to each item and question using a three-point Likert scale, depicted in Table 2 regarding each designed game and the index, labeled as "necessity" [97].

After collecting the data, CVR coefficients were calculated using the following formula:

$$CVR = (n_E - N/2)/(N/2)$$

In this formula, (n_E) represents the number of experts who chose the "necessary" option, and (N) represents the total number of experts who participated in the evaluation. Lawshe [97] presented a table of minimal CVR values based on a test with a 0.05 significance level test to compute the CVR based on the number of experts who assessed the protocol, and recommended that optimal CVR values be at least. Table 3 lists the minimum CVR values for a different number of panel members to decide on.

3.3.2. Content validity index (CVI)

The CVI measures the extent of expert judgments regarding the validity or applicability of the test template or final instrument. It is utilized to determine overall expert agreement and consensus on specific attributes of the program, as well as the importance assigned to its features [98]. Experts employed a four-point Likert scale based on the concepts of "relevance," "clarity," and "simplicity" to assess each item by this index, which the details are explained in Table 4.

The CVI coefficient was subsequently computed using the gathered data according to the following formula [99]:

Table 1

Professional qualifications of the experts.

Field of Expertise	No. of Experts	Qualification and Designation	Average age (Years)	Average Academic Experience (Years)	Average Clinical Experience (Years)
Psychology and Education	9	University Lecturer (Ph.D.)	37	11	7
Computer Software Engineering, Human-Computer Interaction (HCI)	4	Software Engineer, University Lecturer (Ph.D.)	33	12	—
Psychometry	3	University Lecturer (Ph.D.)	43	21	—

Table 2

The Inquiry Employed for the CVR Assessment.

Kindly indicate the level of necessity for including this exercise (game) in a computer-based rehabilitation program.			
Necessity	It is necessary (1)	relatively necessary with the need for review (2)	It is not necessary (3)

$$CVI = N_s / N$$

In this equation, (**Ns**) represents the number of specialists who chose options 3 and 4, while (**N**) denotes the total number of specialists involved in the study. If the CVI score fell below 0.70, the item was considered unacceptable and eliminated. For scores ranging from 0.70 to 0.79, the item was regarded as questionable and required revision. Finally, if the CVI score exceeded 0.79, it was considered satisfactory; in other words, a CVI score of 0.79 was considered the acceptable threshold [100].

4. Findings

4.1. Computer-based cognitive rehabilitation programs in Iran for learning disabilities

This research attempted to design and develop a computer-based cognitive rehabilitation program by reviewing and extracting vital information from previous studies on rehabilitation programs to strengthen the cognitive functioning of children with dyslexia. In the investigation conducted by [101], it was found that computer-based cognitive rehabilitation tasks offer enhanced software features such as immediate assistance, performance feedback, user-controlled exercise management sections (for adjusting difficulty levels or other parameters), and the capability to modify exercise difficulty uniformly or not. Before creating the initial modules of our computer-based cognitive rehabilitation program, we examined several cognitive training programs that are used in Iranian research and target groups, focusing on crucial components including package information, target cognitive functions, and session and exercise specifications, as illustrated in Table 5.

A total of 18 academic papers on cognitive intervention programs developed and used in Iran were examined, exploring consistent patterns derived from previous research findings. These insights serve as valuable guidance for the design and effectiveness of such interventions, enhancing understanding of their mechanisms and practical applications in the field of cognitive rehabilitation. The significance of efficient cognitive processing in cognitive rehabilitation interventions is emphasized by information processing, often associated with working memory and attention [102], cited nine times, representing 50 % of the studies. In approximately 45 % of the studies examined, there was a notable emphasis on working memory, attention and problem-solving

skills, and higher-order cognitive processes showing their significance as fundamental components to consider in the development of a cognitive intervention program.

While, working memory plays a vital role in enhancing various cognitive functions, including tasks such as information storage, transfer, updating, and control [103], the program's flexibility and adaptability could be enhanced by its significant focus on attention and problem-solving skills [104]. Programs targeting reading performance and arithmetic skills exhibit a specialized focus on specific cognitive domains [105], referenced twice, representing only 11 percent of studies. The arrangement and frequency of these cognitive components within the programs offer a comprehensive approach to enhancing cognitive function, highlighting various elements of cognition to promote overall cognitive well-being in the process of program development. The studies, also employed diverse session formats and activity combinations to develop cognitive intervention programs, offering valuable insights into creating successful cognitive rehabilitation programs. These programs typically consisted of between 11 and 20 sessions, each lasting 30 to 60 min for gameplay. In total, approximately 56 % of the games were structured with 45-minute sessions, while the remaining 44 % had longer sessions lasting more than 45 min each.

Incorporating a diverse range of exercises, games, and activities across various sessions indicates a consideration of the complexity of cognitive function by these studies. Research has validated that improving the flexibility and adaptability of intervention approaches may involve incorporating a variety of activities within sessions [106]. The diverse range of exercises within each session aligns with recommendations emphasizing the advantages of comprehensive cognitive training. Additionally, the varying frequencies of program sessions acknowledge the necessity for customized intervention approaches to accommodate different cognitive profiles of clients. This holistic approach, grounded in evidence-based practices, underlines the programs' commitment to improving cognitive functions through personalized and multidimensional treatments [107].

Understanding the complexities of cognitive intervention programs involves analyzing several factors, such as difficulty levels, training

Table 4

Questions Related to Determining CVI.

Please specify how much this exercise (game) could be able to assess the predetermined indicators (Relevance, Clarity, Simplicity)				
Relevance	not related (1)	Relatively related to the need for serious review (2)	related to the need for review (3)	completely related (4)
Clarity	not clear (1)	Relatively clear with the need for serious review (2)	clear with the need for review (3)	completely clear (4)
Simplicity	not simple (1)	relatively simple with the need for serious review (2)	simple with the need for review (3)	completely simple (4)

Table 3

Minimum CVR Values Based on the Number of Panel Members Recommended by Lawshe [97].

Number of Panel Members	5	6	7	8	9	10	12	15	20	25	30	40
Minimum Acceptable CVR values	0.99	0.99	0.99	0.85	0.78	0.62	0.59	0.49	0.42	0.37	0.33	0.29

Table 5
Computer-based Cognitive Rehabilitation Programs.

	Package Information	Target Cognitive Functions	Package Details ^a
1	Sound Smart Educational Software [138–143]	Active memory, attention, problem-solving, information processing, mathematical skills	11 programs, 20 sessions (30–45 min), A: Continuous (basic, intermediate, advanced), B: Program
2	Rayaneh Yar Cognitive Rehabilitation [144,145]	Working memory, planning, inhibition, reading performance, attention, organization	10 games, 10 sessions (45 min), A: Continuous (easy and hard), B: Program
3	Attention and Memory Rehabilitation Program (Aram) [146,147]	Working memory (storage, transfer, updating, and controlling information)	8 tasks, 10 sessions (45 min), A: Continuous (ten levels), B: program, user, C: continuous, D: yes
4	Memory Gym Software Collection [82,148]	Thinking, working memory, attention, problem-solving, processing speed	15 sessions, A: Continuous (6 levels), B: program, C: Continuous
5	Cognitive Rehabilitation Tools for Attention and Concentration [149–152]	Executive functions, attention, concentration, signs of impulsivity, working memory	10 tasks, 16 sessions (60 min), A: Continuous, B: program, C: continuous, D: yes
6	Captain's Log Software [153–155]	working memory, attention, processing speed, problem-solving, inhibition, learning skills	2000 programs, 8–16 sessions (40–60 min), A: Continuous, B: Program
7	Computer-based Family-based Training [156]	Reading performance, accuracy, and attention	272 Exercise, 12 sessions, A: Continuous, B: user, C: Overall
8	N-Back Game Software [75]	Working memory	4 tasks, 18 sessions, A: Continuous, B: user
9	RAM Cognitive Rehabilitation Software [143,157–161]	Reading skills, attention, working memory (ability to store, transmit, update, and control information)	3 games, A: Continuous, B: User, C: Continuous
10	Cogniplus Software [162,163]	Response inhibition, spatial coding, attention	
11	Robo Memo Software [162,164]	Working memory training	
12	Working Memory Training Software [40,165–170]	Working memory training: auditory, visual, spatial (stabilization), executive functions	11 sessions (60 min), A: Continuous (Initial selection with user: one to nine levels), B: program, C: continuous, D: yes
13	Enhances Memory And Concentration Software [171]	Memory, attention and focus, accuracy, executive functions, information processing	11 games, 8 sessions (45 min)
14	Hal Masale Yar Educational Software [172]	Problem-solving (especially math)	8 sessions
15	Brain Ware Safari Software [79,156,173,174]	Information processing, working memory, attention, sensory integrity, and thinking ability	20 pieces of training, 24 sessions (45–60 min), C: Overall
16	Computer Educational Software for Working Memory [175,176]	Auditory and visual memory	4 tasks, 20 sessions (30 min)
17	My Brain Games Collection Computer Software [177]	Memory, math skills, fast adaptation, flexibility, speed of information processing	20 Games, A: Continuous, B: Program, C: Overall
18	Cognitive Game Software [178]	Attention, problem-solving, inhibition,	24 sessions (45 min), A: Continuous

Table 5 (continued)

Package Information	Target Cognitive Functions	Package Details ^a
	planning, cognitive flexibility, and impulse control	

^a A: Difficulty Level Change By, B: Control Training By, C: Performance Feedback, D: Instantaneous Help.

control mechanisms, feedback mechanisms, and the provision of immediate task support. The varying difficulty levels, ranging from 2 to 10, underscore the significance of tailoring interventions to each individual's cognitive requirements [108]. Among the examined programs, 67 percent (12 out of 18) incorporated varying levels of difficulty, an essential practice to accommodate individuals with diverse cognitive profiles. Additionally, there was a distinction between user-controlled and researcher-managed protocols regarding training control mechanisms. Data revealed that four programs (22 percent) allowed users to control their training, while eight programs (44 percent) adhered to researcher-controlled protocols.

This finding aligns with existing research emphasizing the importance of developing interventions that balance participant autonomy with standardization [109]. It was found that continuous feedback mechanisms are essential for improving learning outcomes in cognitive intervention programs. Three of the surveyed programs (17 percent) offered comprehensive evaluations customized to each student's performance, while five of the programs (28 percent) offered ongoing feedback. These feedback systems support targeted skill refining as well as metacognitive awareness, which is in line with accepted best practices in cognitive rehabilitation [110]. Furthermore, including immediate task assistance was identified as a vital element in enhancing program effectiveness [111]. This feature, present in only three programs (17 percent), supports literature suggestions by emphasizing its ability to decrease frustration and enhance skill development.

4.2. Establishment of principles for program development

Examining existing literature on computer-based cognitive rehabilitation for children facing learning challenges aided in addressing the primary research inquiries in this investigation. This investigation indicated that computer-based cognitive rehabilitation refers to the training administered via computer games to improve cognitive functions. Indeed, it relates to the concept of brain plasticity or neuroplasticity [112]. In other words, the ability of computer-assisted cognitive rehabilitation to sustain changes during the follow-up phase can be attributed to brain plasticity [113]. These findings were instrumental in advancing the development of a computer-based rehabilitation program. This program involved designing and creating 24 cognitive games tailored to our target group's cognitive functions, aligning them with the criteria and characteristics of established foundational cognitive games from previous studies and rehabilitation packages.

Consequently, preliminary designs were conducted with the assistance and input of experts, categorizing the cognitive game exercises into three distinct groups: executive functions, memory, and attention. It was expected that each session would involve the weekly practice of a combination of cognitive components associated with various objectives. Puzzle games stand out as a highly favored category capable of fostering enhancement [114], this program, attempts to adopt a broad design framework inspired by puzzle design templates, incorporating elements such as concepts, shapes, and cards into these games. During the program development phase, before engaging in game-based activities, participants are presented with a problem to solve and encouraged to select the optimal solutions. Therefore, following the initial program development, it underwent examination by specialists to assess its

validity and structure.

4.3. Evaluation by proficient experts

4.3.1. Identifying cognitive functions associated with tasks

The study consisted of 16 proficient experts in the fields related to cognitive rehabilitation program development and psychology to evaluate the content validity of the program. These experts participated by completing the evaluation inventory and assessing the program's alignment with its objectives. Following the methodology outlined, data were gathered and analyzed to assess program validity. Table 6 illustrates the percentage distribution of average scores assigned by the experts to each game component within the program. Fig. 3 depicts the distribution of games among various cognitive functions and groups, indicating a nearly uniform distribution across the four cognitive categories.

4.3.2. Determining program's validity

For determining content validity, the CVR and CVI were calculated. Two measures were calculated using the formulas from the methodology section, and the results are displayed in Table 6. According to Lawshe [97], the minimum acceptable value for CVR with 16-panel members is just below 0.49. However, the experts' judgment coefficient was calculated as 0.74 (Table 7), indicating that the developed package is valid. Additionally, the minimum acceptable value for CVI is 0.79 and the Table 6 results show that the program's CVI index for each indicator is above 0.79, confirming the validity of the package. Furthermore, it is worth noting that the validity values for each game were computed and compared against standard scales, revealing that all of them have valid measurements.

4.3.3. Gathering qualitative information regarding additional aspects of the games

In this phase, experts participated in qualitative inquiries concerning the quality of each game individually, and subsequently, adjustments were implemented to the games' framework based on their feedback. Table 8 explains the questions included in the evaluation scale.

4.3.4. Clustering of cognitive functions targeted by game exercises

In this phase, it is needed to select and implement an efficient method for clustering or categorizing the games in the program based on their targeted cognitive function. Cluster analysis, including data mining functions, enables researchers to identify and explore the attributes of each cluster, providing insights into the distribution of their data. Indeed, Clustering divides a dataset into groups, with similar objects falling into or belonging to the same groups. Utilizing distance function constraints, cluster analysis conducts model classifications, distinguishing between different types of objects [115]. This technique categorizes datasets into clusters where datasets within a cluster exhibit greater similarity compared to datasets from different clusters [116].

We explored suitable techniques for data mining and clustering,

ultimately choosing a method called Knowledge Discovery in Databases (KDD). This process involves extracting potentially significant insights, patterns, and correlations from extensive datasets [117]. This process consists of different techniques and tools such as statistics, machine learning, pattern recognition, information retrieval, and other approaches [118]. KDD serves as the foundation for various data analysis techniques, with the k-means clustering technique being a specific method utilized within this framework. Subsequently, to cluster the database, the k-means method uses an iterative strategy. This method partitions n-observations or cases into k-clusters, where each observation belongs to the cluster with the nearest mean.

In data mining and pattern recognition attempts, this unsupervised technique is broadly applied [115]. The clustering process contains steps such as inputting documents, initializing the process, processing the document database, pattern discovery through k-means clustering, and generating the output [119]. To determine the optimal number of clusters, the study employed the Davies-Bouldin (DB) index, which measures cluster separation. This index calculates the ratio of within-cluster and between-cluster distances and determines the average overall clusters. It is chosen for its ease of calculation and its tendency towards a lower score. Additionally, it exclusively utilizes the Euclidean distance function, estimating the distance between the centroids of clusters [120].

Taking this into account, the present study utilized a clustering methodology to categorize and present diverse games according to their efficacy in rehabilitating distinct cognitive functions. Subsequently, this dataset was used to assess the impact of each game on improving specific cognitive functions. Therefore, cognitive functions associated with higher scores were mapped and placed in a relevant bunch of games. For this, the coefficient of variation (CV) of 10 targeted cognitive functions for each game was calculated based on the opinion received from the experts and the appropriate number of cluster's k coefficient, and DB index. The results depicted that the optimum number of clusters was determined as k = 3 with a DB = -0.105. Rapid Miner software was used for all clustering calculations. The average content validity scores for the cognitive functions of the games grouped within each cluster were computed. Then, using the minimum content validity scores as benchmarks, coefficients representing the primary cognitive functions within each cluster were determined and arranged accordingly. Table 9 illustrates the ultimate clustering of the games featured in the cognitive rehabilitation program based on their primary cognitive functions.

The findings indicated that Executive Functions (including visual-scan perception and processing speed reasoning), Attention functions (both focused and divided), and Memory functions (encompassing working memory and visual short-term memory) were closely associated with the games' objectives. Fig. 4 demonstrates notable variations between distinct clusters, highlighting their non-identical nature. As per this diagram, cluster 1 enhances executive function, cluster 2 influences memory function, and cluster 3 improves attention.

Table 6

The percent distribution of average assigned scores to each component of cognitive functions in the overall program (including 24 games).

Cognitive Functions		The extent of average assigned scores (Percent)				
		Very Weak (1)	Weak (2)	Medium (3)	Strong (4)	Very Strong (5)
Memory	Short Term Memory	10	11	24	29	26
	Visual Short-Term Memory	7	9	21	33	30
	Working Memory	6	11	21	33	31
Language	Naming	1	1	23	39	27
Executive Functions	Coordination (Response Time)	17	16	21	26	20
	Perception (Visual Scan)	6	21	21	34	18
	Reasoning (Planning)	1	7	24	39	29
	Reasoning (Processing Speed)	12	15	23	29	20
Attention	Divided Attention	5	14	21	36	25
	Focused Attention	2	11	21	41	24

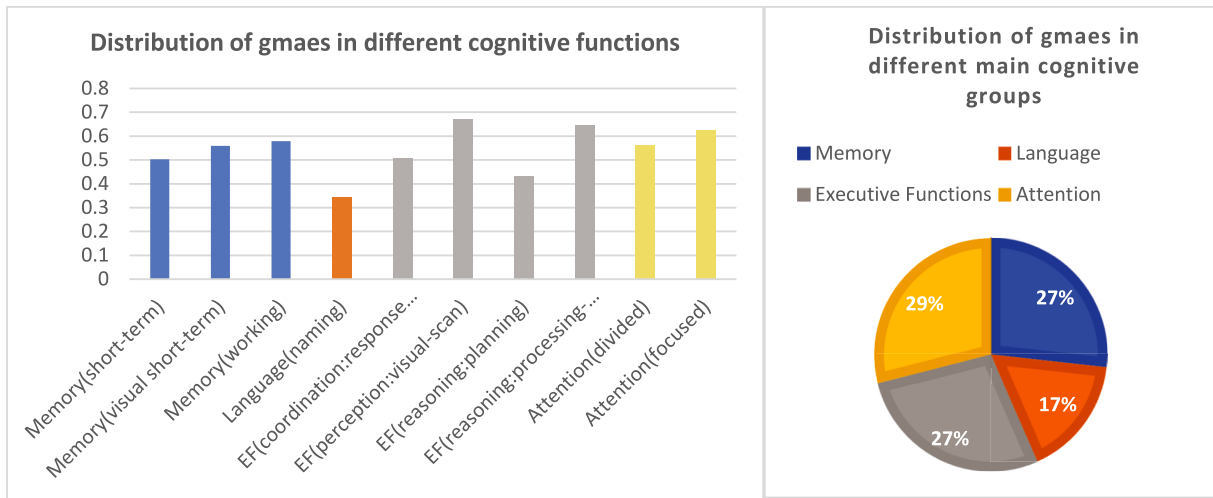


Fig. 3. Experts' assessment of game distribution across various cognitive functions.

Table 7

Evaluation of content validity measures for the developed program based on expert opinions (n = 16).

Number of expert panel members	CVR Necessity	CVI Relevance	Clarity	Simplicity
16	0.74	0.93	0.97	0.94

Table 8

Questions related to designing training.

Please answer the following questions for each game	
1	How many seconds do you recommend for the timing part of the workout, if time permits?
2	At what point of progress do you believe the audience can advance to the next level of difficulty?
3	Do you believe the audience can move to the next level of difficulty with just a few steps after achieving the minimum score?
4	Are the images on the cards suitable, or do they need to be changed in any particular way?
5	Do you have additional recommendations concerning the design principles and arrangement of each game?

4.4. Final program development

This research investigated various aspects of creating a cognitive rehabilitation program initially intended for empowering children aged 7 to 12 with learning difficulties (Dyslexia). The program's scope could also potentially extend to other age groups where children receive early and accurate dyslexia diagnoses. The cognitive rehabilitation program comprised a minimum of 12 sessions, each lasting between 30 and 45 min. To create these games, around 500 pieces of flashcards featuring the Persian and English alphabet, geometric shapes, numbers, and mathematical symbols were created (each group identified with one unique frame color), shown in Fig. 5. These cards were then regularly utilized in various games based on the level of difficulty and various tasks. Also, these cards served as the game's objective, and each exercise strengthened a cognitive skill by using them. In designing these flashcards for game tasks, the priority was to focus on dyslexia-friendly design concepts, which means it is advised to utilize white/black background/font color combinations [121] or black text on a white background [122]. These combinations provide the highest contrast, which aids learning for dyslexic children. Therefore, this recommendation was noticed when selecting backgrounds for game design.

This program has become tailored for Iranian children because, in

Table 9

Classification of cognitive rehabilitation program games by primary cognitive functions.

Clusters	Game's Number	Main Cognitive Functions
Cluster1	3,8,9,12,13,15	1) Reasoning (Processing Speed) 2) Perception (Visual Scan) 3) Attention (Divided) 4) Attention (Focused) 5) Coordination (Response Time)
Cluster2	2,4,5,7,10,11,14,16,17,18,19,20,21,22,23,24	1) Perception (Visual Scan) 2) Memory (Short-Term Memory) – Visual 3) Memory (Working Memory) 4) Memory (Short-Term Memory) 5) Reasoning (Processing Speed)
Cluster3	1,6	1) Attention (Focused) 2) Reasoning (Processing Speed) 3) Perception (Visual Scan) 4) Coordination (Response Time) 5) Attention (Divided)

addition to English alphabets, Persian alphabets were used while designing the tasks. In addition, the initial stage of the program development included frequently used Persian words and phrases that these children were familiar with in their elementary school, which means the core concepts were gathered from their books and curriculums. It has also been tried to use combinations of words together in the design of games, taking into account the expert's considerations, children with dyslexia have more challenges in decoding and word identification. According to Rello et al. [123], the dyslexic participants read noticeably faster when longer words were used, but they understood the text better when shorter words were used. Furthermore, Errors in reading can take many different forms, including missing words, word substitutions, word additions, word inversions, and word mixing within sentences. It seems that youngsters with reading deficiencies have more difficulty remembering meaningless words (non-words) [124]. Therefore, when

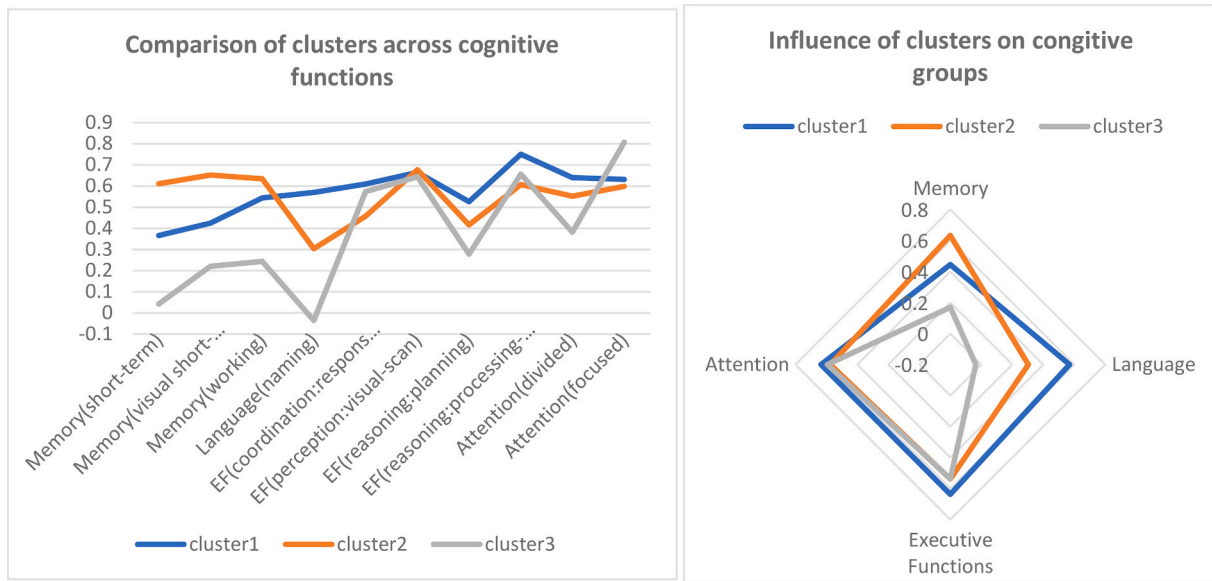


Fig. 4. Status of different clusters across cognitive functions and groups.



Fig. 5. Flashcards used in the program's game-tasks.

developing the game's tasks, an effort was made to include words and phrases in various forms and difficulty levels. Along with the Persian language section, concepts related to shapes, colors, and numbers were also integrated, which can be customized and beneficial for preschool children.

Considering the intervention program, the initial session was designed to familiarize participants with the program's structure, task routines, and phases. During this session, therapists can guide participants through each step and provide instructions using a computer to ensure that any difficulty in-game execution is not due to misunderstanding the instructions. Additionally, the program was primarily developed and organized to encompass 24 game exercises focusing on various cognitive function objectives. Each exercise was structured across four difficulty levels, with approximately 15–20 games per level. This configuration translated to approximately 80 games per exercise. As a result, participants had access to more than 2000 games for their exercise routines. It is important to highlight technical details regarding developing the computer-based program, which followed an interactive web-based approach in which PHP and MySQL programming was used on the server side, alongside HTML, CSS, JavaScript, jQuery, Bootstrap, and Ajax programming were employed on the client side.

Based on the game design mechanism, the game images and options were regularly changed to avoid repetition. In each game, participants received immediate feedback on their performance, helping them to select the correct or incorrect option, along with displaying the correct answer. Eight games, each targeting different cognitive objectives, were

played for an average duration of 30–45 min in every training session. These sessions, expected to span 12 weeks with two sessions per week, primarily focused on enhancing executive functions (including visual-scan perception and processing speed reasoning), attention (both focused and divided), and memory (covering working memory and visual short-term memory). It's crucial to point out that each child's learning journey is unique. They begin at varying levels, progress at different paces, and encounter diverse challenges along the way. To tackle this issue, the intervention program established a structured learning environment, starting with foundational knowledge (scaffolding) before refining the learning process. This approach aimed to integrate engaging topics at the child's comprehension level, while also allowing for customization of the learning experience. The complexity of activities was tailored to suit each child's circumstances, addressing specific issues as they arise.

The initial plan was expected to become progressively more challenging each week. To achieve this, three difficulty levels were assigned for each stage. Participants could advance through these levels based on their progress, with the difficulty increasing as they achieved better scores. The program would then adjust its settings accordingly to accommodate the rate of progress. Additionally, participants could view their activity scores and the game title during gameplay. At the end of each session, participants received a reward as a reinforcement incentive, based on their performance and success in implementing the program. A summary of the instructions for the proposed cognitive rehabilitation training in this program is provided in Table 10.

Table 10

Overview of instructions for describing the proposed cognitive rehabilitation training games in this program.

	Instructions for Training Games	The Five Most Related Functions ^a	Cluster
1	Find the different shapes in the shortest time	10, 6, 8, 5, 9	cluster3
2	Count color blocks due to rules	6, 8, 3, 9, 1	cluster2
3	Find the requested shape among all	9, 6, 8, 4, 10	cluster1
4	Find different color sequences among all	6, 8, 2, 3, 10	cluster2
5	Find a matched number and shape	8, 6, 9, 10, 3	cluster2
6	Click on the circle quickly	10, 8, 6, 5, 9	cluster3
7	Match similar shapes	6, 8, 9, 2, 10	cluster2
8	Choose the most repetitive color	8, 6, 9, 10, 5	cluster1
9	Choose shapes regarding the requested characteristic	6, 8, 3, 9, 4	cluster1
10	Remember the same shapes and then match them	6, 9, 8, 2, 10	cluster2
11	Click on the answer button if the color of the border and the desired shape match	10, 8, 6, 3, 9	cluster2
12	Sort the numbers with the requested order quickly	8, 10, 6, 9, 5	cluster1
13	Click on the answer button according to the rules of displayed color	8, 7, 10, 5, 3	cluster1
14	Remember the colored cells of the table and specify them	6, 2, 8, 1, 3	cluster2
15	Recognize matching of color and displayed color name	8, 5, 4, 10, 9	cluster1
16	Remember the directions and identify the final location of the fish on the table	10, 6, 3, 2, 8	cluster2
17	Remember the shapes and then choose a different shape with them	6, 2, 1, 8, 10	cluster2
18	Remember the initial shape and then select it from the other shapes	6, 10, 2, 1, 8	cluster2
19	Display the shapes in order on the screen and select whether the shape is duplicate or not	1, 2, 3, 6, 9	cluster2
20	A shape is displayed individually. Select whether this shape is the same as the previous shape or not	3, 2, 1, 6, 9	cluster2
21	Several shapes are displayed first and then asked to choose which of the following shapes have been seen or not	2, 1, 3, 10, 6	cluster2
22	Some shapes are displayed then the shapes with blue frames should be selected in the same order displayed and the rest of the shapes have to be ignored.	3, 6, 2, 1, 9	cluster2
23	Several shapes are displayed then the shapes seen must be selected in the same order displayed	2, 3, 1, 6, 5	cluster2
24	Several individual shapes are displayed then the shapes seen must be selected in the same order displayed	2, 3, 1, 10, 6	cluster2

^a 1: Memory (Short-Term), 2: Memory (Visual Short-Term), 3: Memory (Working), 4: Language (Naming), 5: EF (Coordination: Response-Time), 6: EF (Perception: Visual-Scan), 7: EF (Reasoning: Planning), 8: EF (Reasoning: Processing-Speed), 9: Attention (Divided), 10: Attention (Focused).

Finally, for ease of use, the program was entitled “MOBIN Computer-Based Cognitive Rehabilitation (M-CBCR).” Some samples of games are depicted in Fig. 6.

5. Discussion

The goal of this research was to design and develop a computer-based cognitive rehabilitation program using a descriptive method. This program includes various training modules designed to enhance different aspects of cognitive function disorders, specifically tailored to meet the needs of dyslexic children. The present study involved several sequential research steps. Initially, the focus was on identifying the weakest cognitive functions of children with SLD, investigating the effectiveness of cognitive training program aspects, understanding the features of computer-based cognitive rehabilitation programs, and grasping the outcomes of such rehabilitation efforts. Subsequently, the

initial design of the program was formulated based on the insights gained from the first stage. The planned program was developed further, building upon the information acquired in the initial stages. Furthermore, the framework of the program was established, encompassing its objectives, principles, the development of training materials, and software activities. Finally, the program underwent expert evaluation, which included qualitative assessment, quantitative analysis, and verification of content validity by experts.

Recently, the advantageous effects of cognitive training on cognitive abilities in children with dyslexia have been approved by many researchers [37,82,125]. Cognitive rehabilitation intervention methods encompass traditional programs conducted by therapists and computer-based programs. Computer-based interventions, pioneered by Glisky et al. [126] for memory training, represent a novel approach to emerging technologies. Endorsed by expert groups, these programs aim to remediate cognitive deficits in various domains, including attention, memory, executive function, social cognition, and metacognition. The effectiveness of cognitive remediation lies in its application of scientific learning principles, aiming to enhance functional outcomes and everyday functioning in both formal and informal settings [127,128].

Regarding the theoretical basis for the effectiveness of computer-based cognitive rehabilitation, it can be pointed out that this type of rehabilitation usually uses repetitive exercises intended to improve performance and cognitive deficits. Although the mechanism of cognitive rehabilitation performance is still debated, one possibility is that repeated exercise causes neural flexibility in specific neural networks underlying cognitive function [129,130]. In addition, computer-based cognitive rehabilitation offers various advantages compared to other rehabilitation methods. These include cost-effectiveness, as they do not require extensive human resources and are typically more economical. They also offer increased accessibility and the capacity to customize the content, as computers can store and analyze performance data [131].

Within this framework, the program can tailor task difficulty according to users' specific strengths and weaknesses, while also offering objective feedback. This feature is crucial for sustaining user motivation. Moreover, the program offers time flexibility, allowing users to train as often as needed for up to one hour per day, ensuring the minimal commitment required for cognitive enhancements. Furthermore, the program's portability allows for flexible usage anytime, anywhere, it can be used by a wider range of people since it is a tablet-based method and uses a touchscreen system. Thus, not only children with cognitive impairments but also those with simultaneous cognitive and motor disorders can derive benefits from it. In addition, it encompasses multidisciplinary domains, often addressing all cognitive functions, not limited to memory alone.

On the other hand, these computer-based programs have clear considerations, limited educational materials and tips, and meta-cognitive activities that could help sufficiently transfer learning to daily life. In some studies, such as the one conducted by Sala and Gobet [132], it has been found that practicing cognitive training programs at their best state, increases personal performance in tasks similar to the one that has been trained and cannot fully fulfill all individual needed tasks in life. On the other hand, in the analysis of the transition to real and social life, Kosta-Tsolaki et al. [133] found the results of the effectiveness of both computer-based and classical rehabilitation programs for people with cognitive impairments on cognitive functions are satisfactory.

However, it seems that the implementation of a classical cognitive rehabilitation program affects a wider range of cognitive abilities and better transmits cognitive enhancements in real life because one of the major concerns is the generalizability of increasing cognitive components to real-life skills. As diagnosing and intervening for dyslexic children can be complicated, therefore, a comprehensive holistic therapy approach is essential for academic advancement [134]. This approach encompasses individual and group therapies, psychotherapy, psychoeducation, and family therapy, collectively known as Comprehensive Holistic Rehabilitation Programs (CHRP) [135]. Research

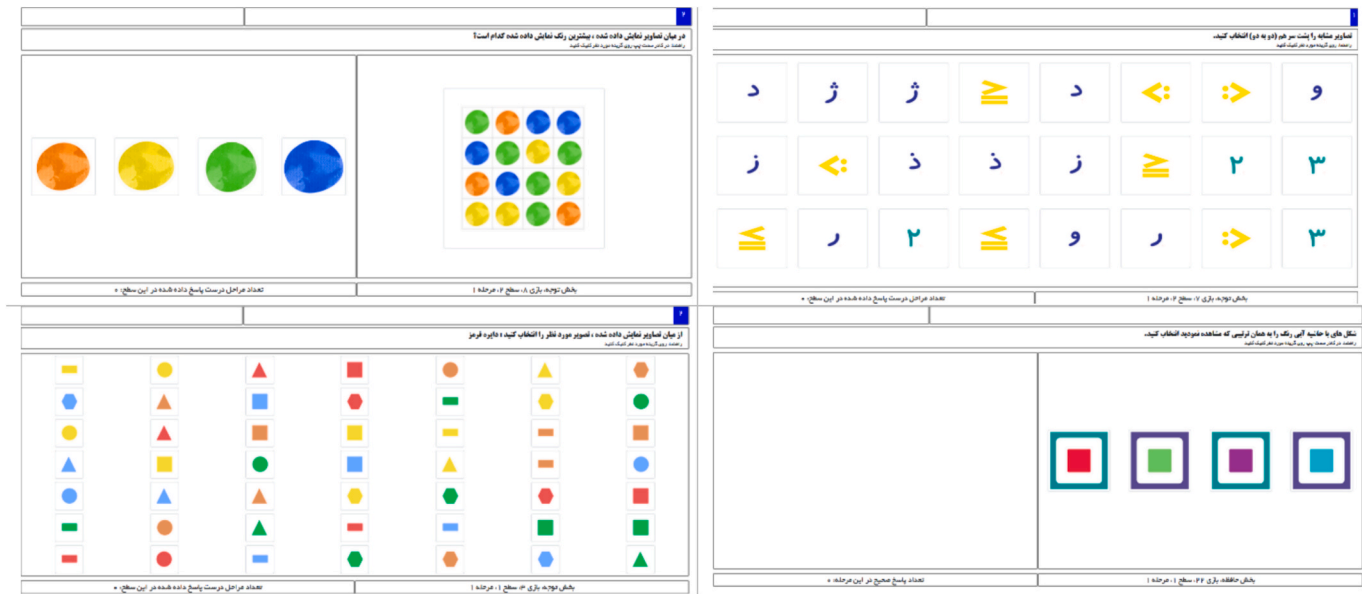


Fig. 6. Some samples of the games in the M-CBCR program.

indicates that patients with cognitive deficits who undergo complete holistic neuropsychological therapy demonstrate better improvements in community functioning compared to those who receive standard rehabilitation [136]. Hence, it is crucial to understand that integrating computer-based programs can enhance the effectiveness of a comprehensive approach.

The results of this study demonstrated the advantages and effectiveness of computer-based rehabilitation. Addressing the potential issue of the limited effectiveness of employing a single rehabilitation approach, Bonk and Graham [137] suggest that blended learning systems offer a more effective solution by integrating face-to-face instruction with computer-mediated instruction. Taking the same concept into account, we define blended cognitive training as the integration of face-to-face group training and computer-based e-training to enhance the effectiveness of a rehabilitation program. The findings indicate that incorporating training software into face-to-face rehabilitation sessions is beneficial.

This allows facilitators to ensure that individuals rapidly understand how to use the technology, processes, and interactions, while also demonstrating the characteristics of various cognitive areas through games. Furthermore, users benefit from the flexibility offered by computer-based programs. As stated earlier, the goal of cognitive rehabilitation is to enhance individuals' capacity to understand and process information, as well as their ability to perform cognitive tasks. However, it's crucial to recognize that cognitive rehabilitation alone may not suffice as a standalone treatment for individuals with cognitive deficits. Research suggests that when cognitive rehabilitation is integrated into a multidisciplinary or interdisciplinary approach, its effectiveness is significantly enhanced [136].

Software development serves as the foundation of computer-based cognitive training, which aims to engage and exercise cognitive functions. While some programs focus on specific cognitive domains, others target a range of domains. In our program, efforts have been made to tailor exercises to diverse cognitive functions. Our program offers several advantages: it allows for the adjustment of difficulty levels based on the user's cognitive abilities and provides prompt and objective feedback, thus reducing the need for subjective therapist intervention. Moreover, performance data is stored in a database, facilitating efficient retrieval, analysis, and management of cognitive rehabilitation processes.

Featuring an interactive interface with sounds and animations, our

program enhances children's attention to tasks, making it more appealing and user-friendly, especially for younger users. It can be utilized on tablet computers with touch screen systems, extending its benefits not only to children with cognitive impairments like dyslexia but also to those with both cognitive and motor impairments. Additionally, it can be utilized by older individuals with severe cognitive impairments. Finally, our program is highly structured and individualized, catering to the specific needs of each user.

The underlying principles and methods of the computer-based cognitive rehabilitation program could potentially be adapted and utilized in different linguistic and cultural settings, as it was created based on an extensive review of global literature and customized to meet the needs of dyslexic children in Farsi-speaking countries. Furthermore, the framework and methodology employed in this study which includes expert reviews and an iterative design process offer a methodical approach that may be utilized to develop programs for dyslexic children in various languages and geographical areas. Additionally, the benefits of computer-based programs including affordability, accessibility, customization, and portability indicate that they may be beneficial and successful in a range of cognitive rehabilitation treatments aimed at diverse cognitive domains.

The study's findings highlight the value of incorporating computer-based cognitive training into all-encompassing rehabilitation programs that incorporate a range of interventions, including family therapy, psychoeducation, individual and group therapy, and psychotherapy. However, The study encountered some limitations, including challenges associated with conducting face-to-face interviews with experts due to the COVID-19 pandemic. Additionally, logistical difficulties arose from the geographical dispersion of professionals, making it challenging to reach and interview them. Another limitation stemmed from limited access to detailed specifications of previously developed computer-based intervention programs, thereby restricting insights into prior work. Furthermore, clinics with minimal experience in implementing computer-based intervention solutions for dyslexia struggled to provide adequate assistance in developing program modules. The lack of national experience in this field also hindered the development of initial modules.

6. Conclusion

The current study developed and evaluated a computer-based

cognitive training program for dyslexic children using a three-step approach to design, develop, and assess the program. The program includes a control mechanism for adjusting level difficulties, allowing for user adaptability and, thereby, optimizing learning processes. The computer-based program was found to be responsive to the cognitive needs of dyslexic children, as perceived by the experts. The findings of the first evaluation provided a promising foundation for further investigation. The study concludes that additional progress in this domain is required to provide more scientific evidence for the use of computer-based training in the treatment process of dyslexia. This research recommends two stages to be added to any newly developed program for the rehabilitation of dyslexic people including pilot intervention and program evaluation by a marketing expert. While the computer-based cognitive rehabilitation program for children with dyslexia seems promising, further research is needed to verify its efficacy and appropriateness for different populations and circumstances. Pilot testing and social networking could be part of future research to enhance the program's availability and utility.

CRedit authorship contribution statement

Faezeh Shabanali Fami: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ali Akbar Arjmandnia:** Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Hadi Moradisabzevar:** Supervision, Software, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The study followed the ethical standards outlined in the 1964 Helsinki Declaration and obtained ethical approval (ID IR.UT.PSYEDU.REC.1400.059) from the University of Tehran and the Iran National Committee for Ethics in Biomedical Research. The participants granted oral informed consent, with due regard for their health, rights, and privacy.

Data availability

Data will be made available on request.

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